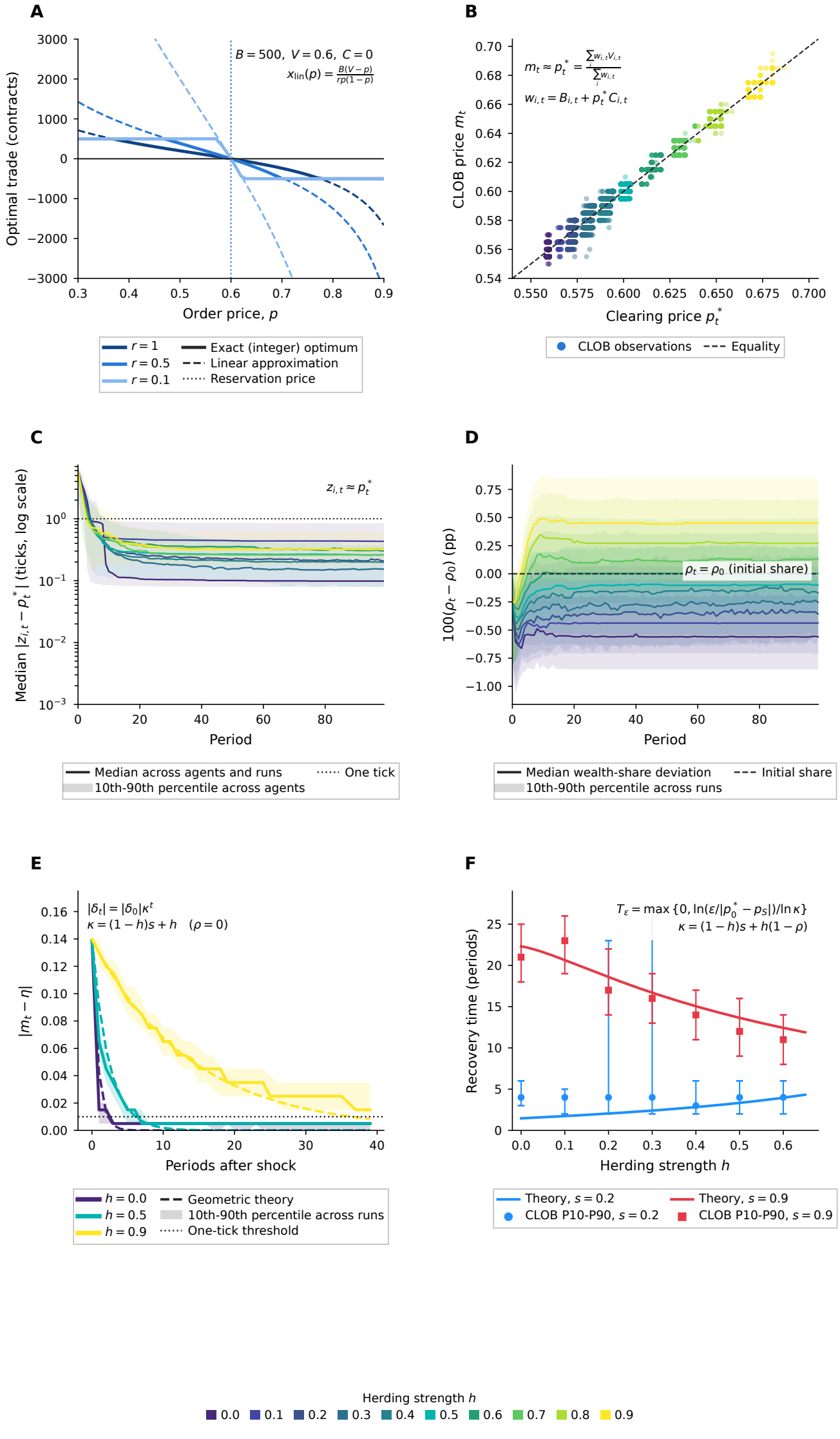

A
B = 500, V = 0.6, C = 0
$x_{\text{lin}}(p) = \frac{B(V-p)}{rp(1-p)}$
Optimal trade (contracts)
Order price, p
r = 1
r = 0.5
r = 0.1
Exact (integer) optimum
Linear approximation
Reservation price
B
$m_t \approx p_t^* = \frac{\sum_i w_{i,t} V_{i,t}}{\sum_i w_{i,t}}$
$w_{i,t} = B_{i,t} + p_t^* C_{i,t}$
CLOB price m_t
Clearing price p_t^*
CLOB observations
Equality
C
$z_{i,t} \approx p_t^*$
Median $|z_{i,t} - p_t^*|$ (ticks, log scale)
Period
Median across agents and runs
One tick
10th-90th percentile across agents
D
$\rho_t = \rho_0$ (initial share)
$100(\rho_t - \rho_0)$ (pp)
Period
Median wealth-share deviation
Initial share
10th-90th percentile across runs
E
$|\delta_t| = |\delta_0|\kappa^t$
$\kappa = (1-h)s + h \quad (\rho = 0)$
$|m_t - \eta|$
Periods after shock
h = 0.0
h = 0.5
h = 0.9
Geometric theory
10th-90th percentile across runs
One-tick threshold
F
$T_\varepsilon = \max\{0, \ln(\varepsilon/|p_0^* - p_S|)/\ln\kappa\}$
$\kappa = (1-h)s + h(1-\rho)$
Recovery time (periods)
Herding strength h
Theory, s = 0.2
Theory, s = 0.9
CLOB P10-P90, s = 0.2
CLOB P10-P90, s = 0.9
Herding strength h
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9